Climbing Simulator in Virtual Reality

SDKs, Engine & Hardware
The application was developed in Unity using the XR Interaction Toolkit and the Oculus XR Plugin, targeting Meta Quest 2. It runs as a Windows PC application with the headset connected via Meta Link. Input is handled through Unity's Input System, and the project also makes use of Microsoft's MRTK library for the in-world virtual keyboard and Meta's official Oculus Hand Models for hand rendering. The scene was built on Universal Render Pipeline (URP) to keep performance stable on the headset.

Key Features

- **Climbing:** Custom systems extend XR Toolkit with two-handed movement, snap-to-grip placement, and two grip types (Trigger = horizontal, Grip = vertical).
- **Stamina:** Per-hand stamina drains while gripping and regenerates when released. Haptics warn at low levels; full depletion forces release (Pro mode).
- **Jumping:** Releasing grips applies a velocity-based impulse; a custom system handles gravity, landing, and falling.
- **Locomotion:** Smooth movement with continuous or snap (45°) rotation, configurable to reduce motion sickness.
- **Procedural Routes:** A generator assembles hold blocks (3 difficulties × 4 directions).
- **UI:** World-space menu (settings, leaderboard), wrist HUD (stamina, timer), and MRTK keyboard for name input.

Testing & Deployment
The application was deployed as a Windows build and tested on a Meta Quest 2 connected via Meta Link. The testing was done on 10 participants who went through a structured test covering all difficulty levels in both Casual and Pro modes. None reported motion sickness, average ratings fell between 4.0 and 4.7 out of 5.

Challenges & Solutions

- **Two-handed climbing:**

  The biggest technical challenge was enabling true two-handed climbing, since the stock Climb Provider only supports one active interactor at a time. This was resolved by maintaining a list of all currently active interactors and averaging their movement deltas each frame.

- **Balancing difficulty**:

  The procedural generator occasionally produced routes that felt harder or easier than the chosen difficulty because off-target blocks could be selected.

Links
Build: Google Drive
Video: YouTube
Source code: github.com/veronikamanhalova/BP